

... Advancing Excellence Through Quality Education ...
KATHMANDU MODEL SECONDARY SCHOOL
 SECOND TERM EXAMINATION 2082

Class: XI 'Science'

Time: 3hrs.

Subject: Chemistry [3011]

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable.
 The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'B' [8x5=40]

SHORT ANSWER QUESTIONS.

1. a) Show with example that all reactions are not redox reaction. [1]
 b) Balance the following redox reaction either by oxidation number method or by ion electron method. [4]

$$C_{12}H_{22}O_{11} + HNO_3 \longrightarrow H_2C_2O_4 + NO_2 + H_2O$$
2. a) State Faraday's First Law of Electrolysis. [1]
 b) Define 1 Faraday. [1]
 c) Iron slab having dimension 50 cm x 30 cm x 20 cm was electroplated all around the surface with a uniform thickness of copper by passing 5 amperes of current for 2 hours in $CuSO_4$ solution. Find the thickness of Cu deposited. Given, the specific gravity of Cu is 8.9 g/cc and its atomic mass is 63.5 amu. [3]
3. a) Write the limitations of Rutherford's atomic model. [1]
 b) How did Bohr interpreted origin of hydrogen spectra on the basis of his atomic model? [4]
4. a) 390 g of SO_2 were treated with 170g O_2 and 120 g water to form H_2SO_4 with the help of following reaction.

$$2SO_2 + O_2 + 2H_2O \longrightarrow 2H_2SO_4$$

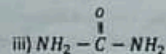
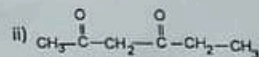
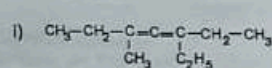
 Find out the limiting reactant and weight in gram and moles of reactants left behind. [3]
 b) What weight of 60% pure H_2SO_4 is required to decompose 25g of Chalk ($CaCO_3$)? [2]
5. a) Differentiate between real gas and ideal gas. Write the condition in which real gas shown ideal behavior. [1+1]
 b) A mixture of ozone and oxygen containing 20% by volume of ozone diffused through a porous plug in 172 seconds, while the same volume of pure oxygen took 164 seconds to diffuse through the same plug. Calculate the relative density and molecular mass of ozone. [3]

6. a) Define electro negativity. [1]
 b) Discuss any four factors that affect electronegativity. [2]
 c) Why is electron affinity of chlorine higher than that of fluorine? [2]
7. a) Write the principle for manufacture of sulphuric acid by contact process. [2]
 b) Give one reaction to show acidic nature of H_2SO_4 . [1]
 c) Compare the bleaching action of SO_2 with Cl_2 . [2]
8. a) Define octane number. A fuel has an octane number 75. What does it mean? [1+1]
 b) Write the reaction for detection of nitrogen as hetero element. [2]
 c) Sodium extract solution must be boiled with HNO_3 before test for halogen, why? [1]

GROUP - 'C' [3x8=24]

LONG ANSWER QUESTION

9. a) State Heisenberg's uncertainty principle. [1]
 b) Why is Bohr's atomic model against the concept of Heisenberg's uncertainty principle? [2]
 c) Write the values of all four quantum numbers for 11th electron of sodium. [2]
 d) Write the differences between orbit and orbital. [3]
10. a) State Avogadro's hypothesis and hence derive the relationship between molecular weight and vapour density. [1+2]
 b) State the law of reciprocal proportion with example. [2]
 c) A hydrocarbon C_xH_y has mass ratio between hydrogen and carbon 1:10.5. One Liter of the hydrocarbon at 127°C and 1 atm pressure weights 2.8g. Find the molecular formula of the hydrocarbon. [3]
11. a) Write the IUPAC name of: [1+1+1]



- b) Define homologous series and write its four characteristics. [1+2]
- c) Define Heterocyclic compounds with example. [1]
- d) Write a chemical reaction to show reducing nature of H_2S . [1]

Best of Luck

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SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET- 'B'

Time: 25 Min.

Subject: Chemistry (3011)

Write the best options among the four options on your answer sheet.

GROUP - 'A' [11×1 = 11]

Multiple Choice Questions

- The set of quantum numbers not applicable for an electron in an atom is
 a) $n=1, l=1, m=1, s=+\frac{1}{2}$ b) $n=1, l=0, m=0, s=+\frac{1}{2}$
 c) $n=1, l=0, m=0, s=-\frac{1}{2}$ d) $n=2, l=0, m=0, s=+\frac{1}{2}$
- If uncertainty in position of an electron is zero, the uncertainty in its momentum would be
 a) Zero b) $> \frac{h}{4\pi}$ c) $< \frac{h}{4\pi}$ d) infinite
- A smuggler could not carry gold by depositing iron on the gold surface since
 a) gold is denser
 b) iron rusts
 c) reduction potential of gold is higher than iron
 d) reduction potential of gold is lower than iron
- $xMnO_4^- + yH_2O_2 \rightarrow 2Mn^{2+} + 5H_2O + 9O_2 + ze^-$ In this reaction, value of x , y and z are
 a) 2, 5, 6 b) 5, 2, 9 c) 3, 5, 5 d) 2, 6, 6
- In Lassaigne's test for detection of sulphur, a violet colour is obtained when sodium nitroprusside is added to sodium extract. It is due to the formation of

- a) $Na_3Fe(CN)_6$ b) $Na_4[Fe(CN)_5NOS]$
 c) $Fe(CNS)_3$ d) $Na_3[Fe(CN)_5NS]$

- Different proportions of oxygen in the various oxides of nitrogen proves the law of
 a) equivalent proportion b) multiple proportion
 c) constant proportion d) conservation of mass
- The pair of gases containing same number of molecules is
 a) 22 g of CO_2 and 72 g of N_2 b) 11 g of CO_2 and 28 gm of N_2
 c) 44 gm of CO_2 and 7 g of N_2 d) 11 g of CO_2 and 7 g of N_2
- If the four tubes of a car are filled to the same pressure with N_2 , O_2 , H_2 and Ne separately, then which one will be filled first?
 a) N_2 b) O_2 c) H_2 d) Ne
- Correct order for second ionization energy is
 a) $Be < C < B < Li$ b) $Be < Li < B < C$
 c) $Li < Be < B < C$ d) $Be > Li < B < C$
- The most stable allotrope of sulphur is
 a) β - sulphur b) α - sulphur
 c) γ - sulphur d) δ - sulphur
- The IUPAC name of the compound $CH_3-\overset{\overset{H}{|}}{\underset{\underset{C_2H_5}{|}}{C}}-\overset{\overset{C_4H_9}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$ is
 a) 2 - butyl - 2 - methyl - 3 - ethylbutane
 b) 2 - ethyl - 3, 3 - dimethylheptane
 c) 3, 4, 4 - trimethylheptane
 d) 3, 4, 4 - trimethyl octane

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